PROJECT REPORT

House Price Prediction Using Linear Regression

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Live demo: https://aryansahu-house-price-prediction.streamlit.app/

1. Project Title

House Price Prediction Using Linear Regression, with a Streamlit web application for live predictions.

2. Introduction

This project is the final assignment of my four-week AI/ML internship. It covers the complete machine learning workflow: loading a dataset, cleaning and analysing it, training a model, evaluating the results and deploying the model as a web application. The system estimates the price of a house from basic property details.

3. Problem Statement

The price of a house depends on several factors at once, such as size, number of rooms, age, plot size, garage and neighborhood quality. This project builds a supervised regression model that predicts a house price from these seven features, and makes it available through a simple web interface.

4. Objectives

- Load, inspect and understand the house price dataset.
- Check and clean the data (data types, missing values, duplicates, outliers).
- Explore relationships between features and price.
- Scale the features and train a Linear Regression model.
- Evaluate the model using R-squared and adjusted R-squared.
- Deploy the model as an interactive Streamlit application.

5. Technologies Used

Tool	Purpose
Python	Main programming language
Pandas, NumPy	Data loading, cleaning and numerical operations
Matplotlib, Seaborn	Data visualisation
Scikit-learn	Scaling, train-test split, model training and evaluation
SciPy	Pearson correlation
Joblib	Saving the trained model and scaler
Jupyter Notebook	Analysis and model development
Streamlit	Web application and deployment

6. Dataset Description

The dataset is Home Value Insights from Kaggle (file: house_price_prediction.csv). It contains 1,000 records and 8 columns: seven input features and one target (House_Price).

Column	Description
Square_Footage	House area in square feet
Num_Bedrooms	Number of bedrooms
Num_Bathrooms	Number of bathrooms
Year_Built	Year of construction (1950–2022)
Lot_Size	Size of the lot
Garage_Size	Garage size value (0–2)
Neighborhood_Quality	Quality score from 1 to 10
House_Price	Target value to predict

7. Data Preprocessing

- Checked data types: all columns were numeric.
- Missing values and duplicate rows: none found.
- Converted values to numeric, flagged impossible values (zero, negative or future years) and filled any gaps with the column median.
- Outlier check using boxplots and the IQR method: no outliers found, so no rows were removed.
- Applied StandardScaler to all seven input features.

8. Exploratory Data Analysis

Histograms showed the values are spread fairly evenly across their ranges. The scatter plot of square footage against price shows a clear straight-line relationship. Pearson correlation with the house price is shown below.

Feature	Correlation with price
Square_Footage	0.9913
Lot_Size	0.1604
Garage_Size	0.0521
Year_Built	0.0520
Num_Bedrooms	0.0146
Num_Bathrooms	-0.0019
Neighborhood_Quality	-0.0078

Square footage is by far the strongest factor in this dataset; the other features have little linear relationship with price.

9. Model Development

A Linear Regression model was chosen because the target is a continuous value and the main relationship is linear. The data was split 80% for training and 20% for testing (random_state = 42). The model was trained on the training set, used to predict the test set, and then saved as house_price_model.pkl along with the scaler (house_price_scaler.pkl) for use in the web app.

10. Model Evaluation

The model was evaluated on the test set using R-squared, which shows how much of the variation in price the model explains, and adjusted R-squared, which accounts for the number of features used.

11. Results

Metric	Score
R-squared	0.998426
Adjusted R-squared	0.998369

The model explains about 99.84% of the variation in house prices on the test data. This result applies to this dataset and this split only, and may not carry over to real-world housing data. Predictions are estimates, not guaranteed market values.

12. Screenshots

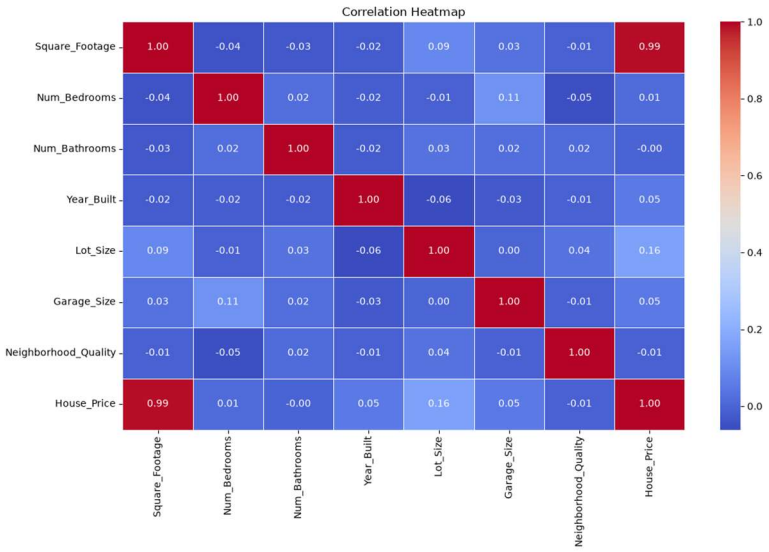


Figure 1: Correlation heatmap

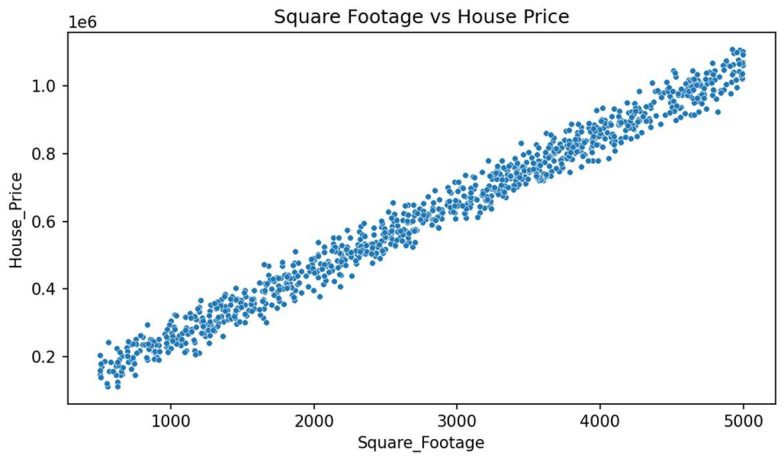


Figure 2: Square footage vs house price

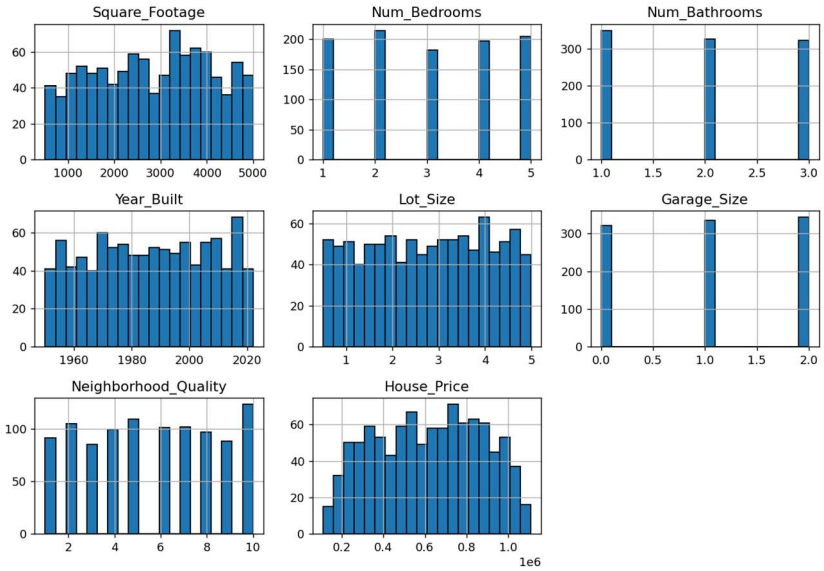


Figure 3: Histograms of numeric columns

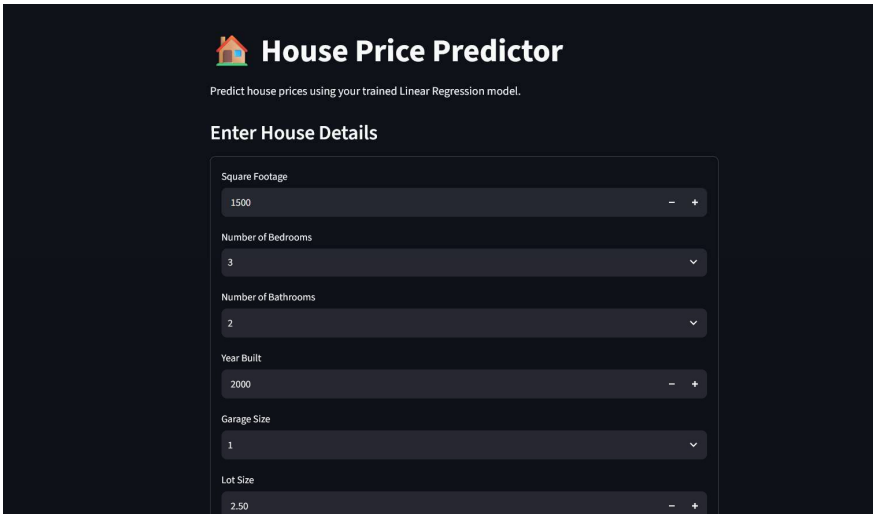


Figure 4: Application input form

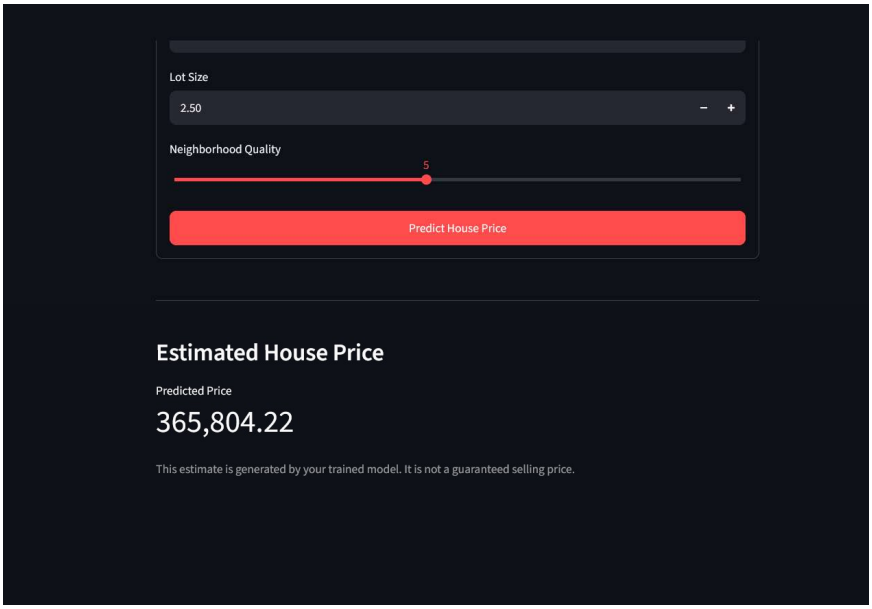


Figure 5: Application showing a predicted price

13. Challenges Faced

- Column order: the scaler and the model were trained with different column orders, so the app reorders the inputs between the two steps to keep predictions correct.
- Very high score: an R-squared near 0.998 was explained by square footage dominating the dataset.
- Weak features: neighborhood quality and bathrooms showed almost no relationship with price.
- Deployment: the model, scaler and app files had to be organised together so the app loads correctly.

14. Future Improvements

- Compare with other models such as Ridge, Random Forest and Gradient Boosting.
- Report MAE, MSE and RMSE, and use cross-validation.
- Fit the scaler on training data only to avoid data leakage.
- Test on data from other locations and time periods.
- Add clearer feature units and stronger input validation in the app.

15. Conclusion

The project delivered a complete machine learning pipeline, from raw data to a deployed web application. The Linear Regression model achieved an R-squared of about 0.998 on the test data, and the Streamlit app makes it easy for anyone to try. The key lesson was that model quality depends heavily on the data, and that small details such as feature order matter when moving from a notebook to a live application.